

MacSync-W-Odor-X1 Wi-Fi Qualitative Odor Sensor Datasheet

Product Overview

MacSync-W-Odor-X1 is a qualitative **Wi-Fi-based odor monitoring sensor** designed to detect, track, and trend **odor intensity levels** in industrial, municipal, and commercial environments where reliable IP connectivity is available. Instead of measuring individual gas concentrations in ppm, the device evaluates overall odor presence and strength, making it ideal for applications where **nuisance odor detection, early warning, and compliance monitoring** are critical.

Built on advanced sensing elements and intelligent signal processing, MacSync-W-Odor-X1 converts complex odor signatures into a **simple odor index / odor level**, enabling easy interpretation and actionable insights. This qualitative approach allows operators to identify odor spikes, persistent odor patterns, and abnormal events without requiring gas-specific calibration or complex analysis.

Designed for continuous operation in real-world environments, the device features an **industrial-grade, ingress-protected enclosure** suitable for **sewage treatment plants (STP), wastewater facilities (WWTP), landfills, waste processing units, factories, commercial buildings, and indoor utility areas** within Wi-Fi coverage.

MacSync-W-Odor-X1 transmits odor data securely over **standard Wi-Fi networks (2.4 GHz / 5 GHz)** directly to **on-premise servers, dashboards, SCADA systems, or cloud platforms** using industry-standard protocols such as **MQTT, HTTP, and HTTPS**—without the need for gateways or long-range wireless infrastructure. **Local BLE-based configuration via the Macnman Maya App** enables fast provisioning, diagnostics, and maintenance.

Whether used for regulatory reporting, community odor complaint management, or proactive operational control, MacSync-W-Odor-X1 delivers a **reliable, low-maintenance, and data-driven Wi-Fi odor monitoring solution**, transforming subjective smell complaints into measurable intelligence.



Key Features

Qualitative Odor Intensity Detection

Measures overall odor presence and strength using a qualitative odor index, ideal for nuisance odor monitoring without ppm-level gas complexity.

Early Warning & Threshold Alerts

Configurable odor thresholds trigger alerts for sudden odor spikes, persistent odor conditions, and abnormal events.

Industrial-Grade & Outdoor-Ready Design

Rugged, ingress-protected enclosure suitable for STP, WWTP, landfills, waste processing plants, factories, and utility areas within Wi-Fi coverage.

Odor Index & Trend Analysis

Converts complex odor signatures into a simple odor level for long-term trend analysis and correlation with operational events.

Wi-Fi Based Connectivity

Direct odor data transmission over standard Wi-Fi networks (2.4 GHz / 5 GHz) without gateways or long-range infrastructure.

No Gas-Specific Calibration Required

Qualitative sensing eliminates the need for individual gas calibration, reducing maintenance effort and operational complexity.

Secure IP-Based Communication

WPA2/WPA3 Wi-Fi security with TLS-encrypted data transmission ensures safe and reliable odor data delivery.

BLE Configuration via Maya App

Local device configuration, diagnostics, and maintenance using the Macnman Maya mobile application.

Suited Applications

Sewage Treatment Plants (STP)

Continuous monitoring of nuisance odors from inlet chambers, aeration tanks, and sludge handling areas within Wi-Fi-enabled facilities.

Landfills & Waste Disposal Sites

Qualitative odor monitoring to identify emission trends and manage odor impact within site premises.

Industrial Manufacturing Plants

Odor detection around chemical processing, food processing, paper mills, and industrial production areas.

Environmental Compliance & Reporting

Supports regulatory monitoring programs and data-driven reporting for odor-related compliance requirements.

Flexible Reporting Modes

Supports time-based, threshold-based, and event-driven reporting to balance responsiveness and power consumption.

On-Prem & Cloud Integration Ready

Compatible with on-premise servers, dashboards, SCADA systems, and cloud platforms using MQTT, HTTP, and HTTPS.

Wastewater Treatment Plants (WWTP)

Odor intensity tracking across treatment processes to detect abnormal events, process inefficiencies, and odor spikes.

Solid Waste & Composting Facilities

Odor level monitoring to identify high-odor zones and optimize waste handling and composting operations.

Commercial Buildings & Utility Areas

Odor monitoring in basements, utility rooms, garbage areas, and common service zones inside buildings.

Community Odor Complaint Correlation

Provides objective odor data to correlate citizen complaints with actual odor events and operational conditions.

Technical Specifications

General Specifications

Part Order Number	MacSync-W-Odor-X1
Product Brand	Macnman Technologies Pvt Ltd

Temperature Measurement

Range	-30°C to +70°C
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Accuracy	0°C to +70°C: $\pm 0.3^{\circ}\text{C}$ & -30°C to 0°C: $\pm 0.6^{\circ}\text{C}$
Resolution	0.1°C
Calibration	Factory calibrated
Probe Options	Internal & External Both Options

Humidity Measurement

Range	0% to 100% RH
Accuracy	10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH
Resolution	0.5% RH
Calibration	Factory calibrated

Pressure Measurement

Range	300 hPa to 1100 hPa
Accuracy	± 1.0 hPa (typical at 25°C, 300–1100 hPa)
Resolution	0.1 hPa
Long-Term Stability	$\leq \pm 1$ hPa/year
Calibration	Factory calibrated

NH3 Measurement

Gas Detected	Ammonia (NH ₃)
Detection Range	1 to 300 ppm
Sensitivity	R_0 (in air) / R_s (in 50 ppm NH ₃) ≥ 3
Measurement Type	Gas concentration (ppm)
Operating Temperature	-10°C to +50°C
Operating Humidity	15% to 90% RH (non-condensing)
Expected Lifespan	≥ 5 years (in air)
Calibration	Factory calibrated

H2S Measurement

Gas Detected	Hydrogen Sulfide (H ₂ S), Benzene
Detection Range	0.5 to 50 ppm
Sensitivity	R _o (in air) / R _s (in 50 ppm H ₂ S) ≥ 3
Measurement Type	Gas concentration (ppm)
Operating Temperature	-10°C to +50°C
Operating Humidity	15% to 90% RH (non-condensing)
Expected Lifespan	≥ 5 years (in air)
Calibration	Factory calibrated

Wireless Communication

Wi-Fi Standards	IEEE 802.11 b/g/n/ac
Frequency Bands	2.4 GHz & 5 GHz (Dual-Band)
Operating Modes	Station Mode (Client) Access Point Mode (Configuration)
Network Protocols	MQTT, HTTP, HTTPS
Security	WPA2-PSK / WPA3 TLS / SSL Encrypted Communication
IP Configuration	DHCP / Static IP
Typical Indoor Range	Up to 100 m (environment dependent)

LED Status

LED Type	1 Internal Multi-Color LED
Solid Blue	Device is connected with Maya App
Flashing Blue	OTA (Over-the-Air) update in progress
Slow Blinking Green	Attempting to connect to the network
Single Red Blink	A confirmed parameter of MacSync is being modified

Device Configuration

Configuration	Via Macnman Maya Android App
Uplink Modes	Time Based / Threshold Based / Sampling Based

Stored Records*	30,000 Records
Remote Configuration	JSON Format Data - from server

Supported Cloud & Platforms

Deployment Models	On-Premise Servers Private Cloud Deployments Public Cloud Platforms
IoT Cloud Platforms	AWS IoT Core Microsoft Azure IoT Hub Google Cloud (via MQTT / HTTPS) Custom IoT Platforms
Integration Methods	MQTT Brokers REST / HTTPS APIs Webhooks
Enterprise Systems	SCADA Systems BMS / IBMS Platforms Data Lakes & Analytics Engines

Physical Characteristics

Power Supply	9-36 V DC
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)
Ingress Protection	IP65 standard (IP67 optional variant)
Housing Material	Glass filled Nylon
Color	Matte White
Weight	~110g
Mounting Options	Pole Mounting / Wall Mounting

Electrical Characteristics

Input Voltage Range	9-36 V DC (External Supply)
Power Consumption	~100 μ A in sleep mode ~150 mA during transmission
Input Impedance	>10 M Ω for voltage inputs

Communication Options

BLE Connectivity	For local configuration and diagnostics via Maya App
Cloud Protocols	MQTT / HTTPS
Serial Interface	RS485 Modbus RTU support (optional)

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

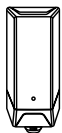
Data Encryption	AES-256 end-to-end encryption
Authentication	OTAA activation modes
Firmware Protection	Secure boot and signed OTA updates

MacSync-W-Odor-X1 Variants

VARIANT NAME	POWER SOURCE	BATTERY CAPACITY	SOLAR SUPPORT	EXTERNAL FLASH	IDEAL FOR
MacSync-W-Odor-X1-BO	Battery	19 Ah			Remote or portable deployments with low power

What's in the Box?

Make sure box contains following items before installations



MacSync



**Mounting
Clip**



**User manual &
Warranty Card**



**Junction
Box***



**Mounting
Screws**

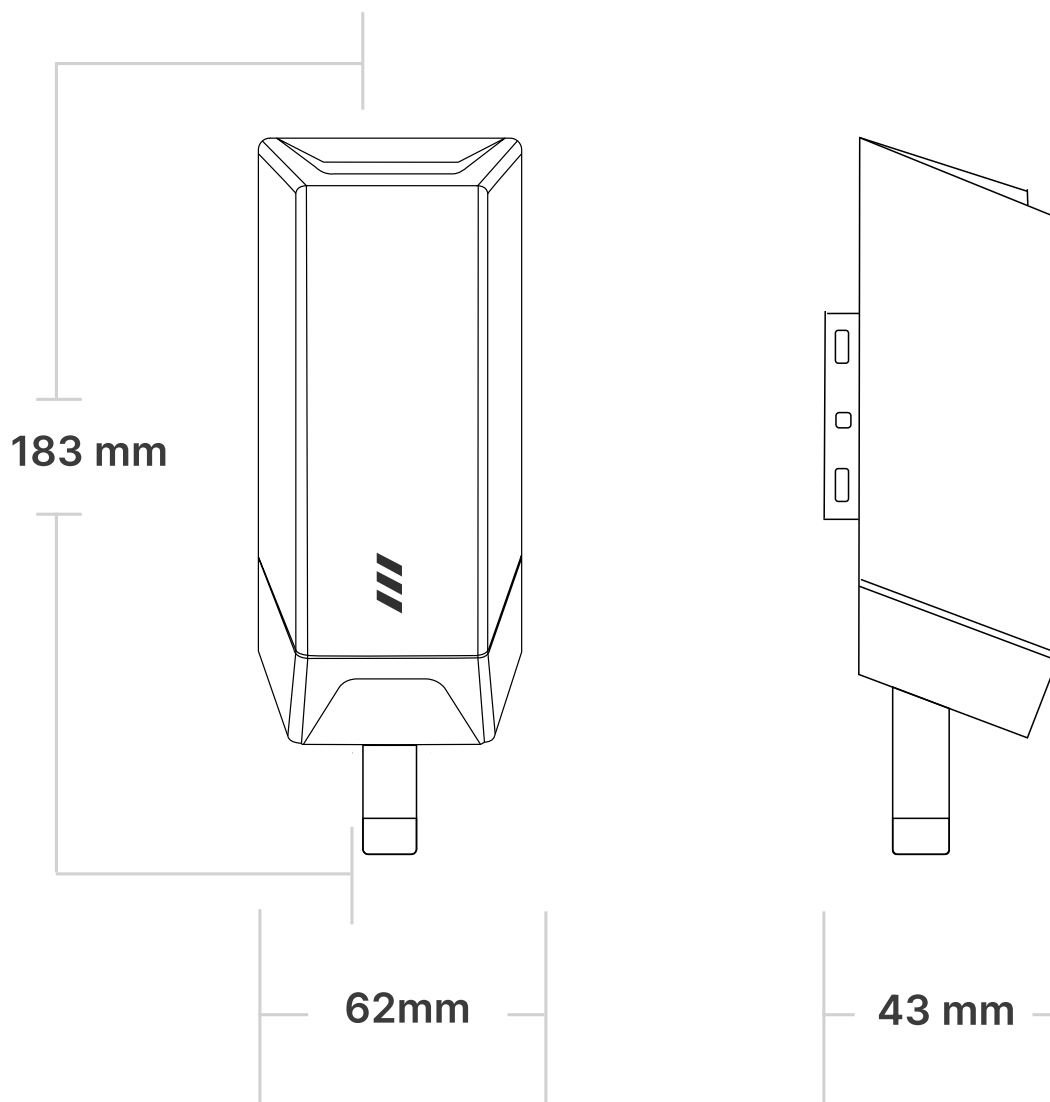


**Power
Adapters***

** Note**

Accessories must be purchased separately & may not be included in the box.

MacSync-W-Odor-X1 Dimensions



Frequently Asked Questions

What is the MacSync-W-Odor-X1 used for?

MacSync-W-Odor-X1 is a qualitative Wi-Fi odor monitoring sensor that detects, tracks and trends odor intensity in industrial and facility environments. Typical applications include Sewage Treatment Plants (STP), Wastewater Treatment Plants (WWTP), Landfills & Waste Disposal Sites, Solid Waste & Composting Facilities, Industrial Manufacturing Plants and Commercial Buildings & Utility Areas.

What does the MacSync-W-Odor-X1 measure, and how accurately?

Measurement specifications of the MacSync-W-Odor-X1:

- Temperature: range -30°C to +70°C, accuracy 0°C to +70°C: $\pm 0.3^\circ\text{C}$ & -30°C to 0°C: $\pm 0.6^\circ\text{C}$, resolution 0.1°C.
- Humidity: range 0% to 100% RH, accuracy 10% to 90% RH: $\pm 3\%$ RH & <10% or >90% RH: $\pm 5\%$ RH, resolution 0.5% RH.
- Pressure: range 300 hPa to 1100 hPa, accuracy ± 1.0 hPa (typical at 25°C, 300–1100 hPa), resolution 0.1 hPa.
- NH₃: range 1 to 300 ppm.
- H₂S: range 0.5 to 50 ppm.

How does the MacSync-W-Odor-X1 connect and send data over Wi-Fi?

The MacSync-W-Odor-X1 connects over IEEE 802.11 b/g/n/ac Wi-Fi on 2.4 GHz & 5 GHz (Dual-Band). Security: WPA2-PSK / WPA3, TLS / SSL Encrypted Communication. IP configuration: DHCP / Static IP. It sends data using MQTT, HTTP, HTTPS. Typical indoor range is up to 100 m (environment dependent).

How is the MacSync-W-Odor-X1 powered?

The MacSync-W-Odor-X1 is powered from 9–36 V DC (External Supply).

What is the IP rating and operating temperature of the MacSync-W-Odor-X1?

The MacSync-W-Odor-X1 has an ingress protection rating of IP65 standard (IP67 optional variant) and an operating temperature range of -40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

How is the MacSync-W-Odor-X1 configured?

The MacSync-W-Odor-X1 is configured with the Macnman Maya Android app. Supported uplink modes: Time Based / Threshold Based / Sampling Based. It can also be configured remotely with JSON data sent from the server.

Which cloud platforms does the MacSync-W-Odor-X1 work with?

The MacSync-W-Odor-X1 can send data to platforms including AWS IoT Core, Microsoft Azure IoT Hub, Google Cloud (via MQTT / HTTPS), Custom IoT Platforms. Deployment models: On-Premise Servers, Private Cloud Deployments, Public Cloud Platforms. Integration methods: MQTT Brokers, REST / HTTPS APIs, Webhooks.

Related Products and Resources

- Related datasheets: [MacSync L Odor X1 LoRaWAN qualitative odor sensor](#), [MacSync-WCO2X Wi-Fi CO₂ indoor air quality sensor](#), [MacSync-WGSX2 Wi-Fi H₂S and NH₃ gas sensor](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)